INCH-POUND

MIL-R-14865(AR)
AMENDMENT 3
30 April 1991
SUPERSEDING
AMENDMENT 2
12 July 1973

MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, LASER: 10559281

This amendment forms a part of MIL-R-14865(AR), dated 2 March 1970, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

Table I, Item 1.12, delete "240" and substitute "1800"

Table I, Item 1.13, delete "620" and substitute "2700"

Table I, add:

"1.18 NAND Gate	1 Gate - DTμL 946 or equivalent	J3-G(return to J3-N)
1.19	62 pf ± 5%	J3-G and J3-N"

PAGE 5

Table II, delete "J4-N" and substitute "J4-W"

PAGE 8

Figure 4, delete "40 ± 32 ms" and substitute "100 ± 60 ms"
* Figure 4, delete "0.6 ± 0.5 s" and substitute "0.8 ± 0.6 s"

PAGE 9

Paragraph 3.3.1.5.1, delete and substitute

"3.3.1.5.1 AGC Inhibit. Under the above conditions and with the receiver-transmitter window covered, the d-c voltage at J3-D shall be 27.0 ± 3.0 Vdc. With the interlock switch (SI) actuator on the Reply Gating Card Assembly (A4-10559295) pressed, the voltage at J3-D shall be 0 ± 1 Vdc."

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.

PAGE 10

Table III, Item 18: Delete "Output" values and substitute:

OUTPUT PINS OF J4

P	R	S	T	p	q
"H	L	L	L	L	H"

PAGE 11

Table III, Note 4, add "or 9990"

PAGE 12

Table V, delete "1.4 ± IV (positive polarity)" and substitute "+4.0 ± 1.5 Vdc"

PAGE 13

Table VII, Item 10 4/, delete output and substitute:

OUTPUT PINS OF J4

Y	p	q	P	R	S	T
"NA	L	H	L	L	L	L"

Table VII, Note 5, add "or 9990"

PAGE 15

Paragraph ""3.3.1.11" renumber to "3.3.1.14"

Add paragraphs 3.3.1.11 thru 3.3.1.13 and Figure 5:

"3.3.1.11 A-trigger threshold. Adjust R-10 on the PFN Charge Power Supply Assembly (P/N 10559345) so that the energy output of the receiver-transmitter is 18.0 ± 3.0 millijoules. The voltage level at J4-b shall be + 4.0 ± 1.6 volts.

3.3.1.12 Laser output timing. When the transmitter is fired, the time between the reset pulse and the A-trigger, at test connector pins J3-B and J3-G respectively, shall be as shown by the waveform of Figure 5.

3.3.1.13 PFN discharge time. With the removal of the 24 Vdc power source of Table I, the PFN current shall decrease to less than 7 microamperes within 25 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.

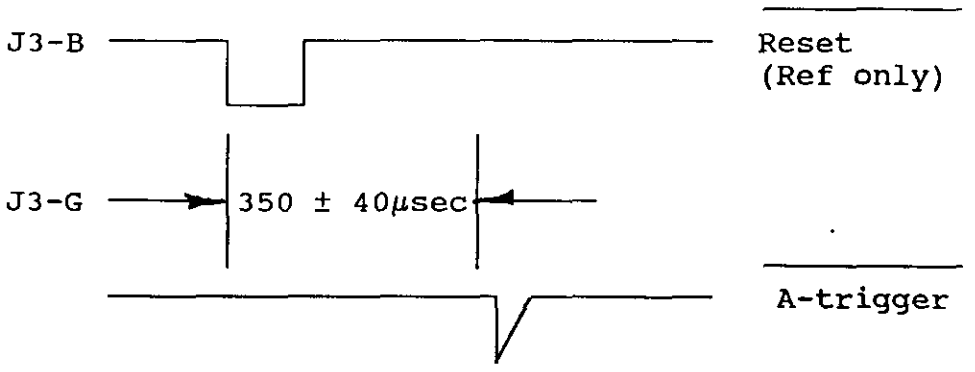


Figure 5. Laser Output Timing

PAGE 15

Paragraph 3.3.2.1, delete "through 3.3.1.10" and substitute "through 3.3.1.13"

Paragraph 3.3.2.2, delete and substitute:

"3.3.2.2 Operating temperatures. The receiver-transmitter shall meet the requirements of 3.3.1.1 through 3.3.1.10, except 3.3.1.1.2, 3.3.1.1.3, 3.3.1.2, and 3.3.1.5, while exposed to and stabilized at -40°F and +125°F, subsequent to which it shall meet the requirements of 3.3.1.1 through 3.3.1.13 at ambient temperature (60°F to 90°F)."

PAGE 16

Paragraph 3.3.2.3 and 3.3.2.4, delete "through 3.3.1.10" and substitute "through 3.3.1.13"

PAGE 17

Paragraph 4.2, add to second sentence "plus three each of all items covered by SQAP."

PAGE 18

Table X, delete item 116 and substitute:

"116.	A-trigger threshold	3.3.1.11	4.6.12
117.	Laser output timing	3.3.1.12	4.6.13
118.	PFN discharge time	3.3.1.13	4.6.14
119.	Receiver-transmitter leakage	3.3.1.14	4.6.15"

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.

PAGE 19

TABLE XII, delete Test Procedure column and substitute:

"TEST PROCEDURE

4.6.16.1
4.6.16.1
4.6.16.2
4.6.16.3
4.6.16.4"

PAGE 21

Table XIII, Items 5 and 6, delete and substitute:

"5.	Optical Power Meter	Instrument capable of measuring peak power of laser beam up to 1 x 10^7 watts at 6943 Å."
"6.	Energy Meter	Instrument capable of measuring energy of 0-100 millijoules."

PAGE 21

Table XIII, Item 11, delete 11, delete "15 percent" and substitute "80 percent"

PAGE 23

Paragraph 4.6.6.1, PFN voltage note, add:

"The PFN voltage shall not exceed 1300 volts."

Paragraph 4.6.6.1, prior to the operation note, add:

"A-trigger threshold. The A-trigger threshold test (4.6.12) shall be performed prior to performing ranging."

Paragraph 4.6.10, add: "Item 5 or 4 and 6 of Table XIII are utilized."

PAGE 24

Paragraph 4.6.11.2, delete "42 ± 1 dB" and substitute "41 ± 1 dB"

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.

PAGES 25 thru 27

Renumber paragraphs:

FROM	TO
"4.6.12	4.6.15
4.6.12.1	4.6.15.1
4.6.13	4.6.16
4.6.13.1	4.6.16.1
4.6.13.2	4.6.16.2
4.6.13.3	4.6.16.3
4.6.13.4	4.6.16.4"

PAGE 25

Add paragraphs:

"4.6.12 A-trigger threshold test. Set the energy output level of the receiver-transmitter to 18 ± 3 millijoules. Verify that the requirement of 3.3.1.11 is met, utilizing Items 2 and 6 of Table XIII.

4.6.13 Laser output timing. With the oscilloscope of Table XIII connected to connector pins J3-B (reset) and J3-G (A-trigger) of the test access connector, the requirement of 3.3.1.12 shall be met.

4.6.14 PFN discharge time. Connect a microammeter between pins A (±) and N(-) on test access connector J3. Remove the 24 Vdc power source (item 2.1) of Table I. The PFN discharge current shall meet the requirements of 3.3.1.13."

Paragraph 4.6.15, delete "3.3.1.11" and substitute "3.3.1.14"

Paragraph 4.6.15.1 (b) delete and substitute:

"(b) After assembly, repressurize the laser head section with an inlet pressure of 14.5 ± 1 psig and allow it to purge for a minimum of 20 minutes. Reduce the internal pressure to ambient. Then repressurize to 7.0 ± 0.5 psig."

Paragraph 4.6.16, delete "and the tolerance requirements for vibration frequency measuring instruments shall not apply."

Paragraph 4.6.16.1, delete "than 7°F" and substitute "than 3°F"; in (d), (h), and (j): delete "through 4.6.11" and substitute "through 4.6.14"; in (d) and (h): delete "4.6.10.2" and substitute "4.6.11.3"; and delete "4.6.3 and 4.6.6." and substitute "4.6.3, 4.6.6, and 4.6.12."

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.

PAGE 26

Paragraph 4.6.16.2(a), delete and substitute:

"(a) Test in accordance with Method 514.1 Procedure I of MIL-STD-810 for category (f) equipment except that resonance search and dwell on resonance shall not be required and the acceleration shall not exceed ± 4g and the maximum excursion (double amplitude) shall be 0.5 inch."

Paragraph 4.6.16.2(e), delete "through 4.6.11" and substitute "through 4.6.14."

PAGE 27

Paragraph 4.6.16.3(c), delete "through 4.6.11" and substitute through 4.6.14."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army - AR

Preparing Activity:
Army - AR

(Project 1240-AB17)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:57Z
Check the source to verify that this is the current version before use.